NON-MF GROUPS

SAUERS

Some groups are not MF. For example, $H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1,2))$ is a simple property-(T) group, and every homomorphism from H to an MF group is trivial. We also construct a two-generated, finitely presented, torsion-free property-(T) group with the same rigidity, and a non-MF affine–Clifford group W that is an extension of $\mathbb{Z}$ by a locally residually finite group. The common mechanism is a normal property-(T) subgroup contained in the normal closure of commutators arising from a one-sided compression. Consequently, $C^*_{\mathrm{r}}(H)$ is separable, stably finite, and not MF, while the canonical trace of W is amenable but not quasidiagonal.

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1. Introduction

The counterexample is

$$H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1,2)),$$

the elementary group of 12×12 matrices over the binary Leavitt algebra. Khanh–Thanh show that H is isomorphic to the unit group of the binary Leavitt algebra [K–T, Proposition 4.2 and Corollary 4.4]. OpenAI used this group to construct the first nonsofic group [OAI, Chapter 3].

A group G is *MF* [CDE] if there are positive integers d_n and maps $V_n\colon G \to \mathcal{U}(d_n)$, with $V_n(1) = 1$, such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G),$$

and

$$\limsup_n \|V_n(g) - 1\| > 0 \qquad (g \in G \setminus \{1\}).$$

In other words, G is MF if there are finite unitary matrices, one for each element of G, that multiply correctly up to an error tending to zero in operator norm while no nonidentity element converges to the identity matrix. For countable G, the separation may equivalently be required along the full sequence with a constant independent of g [Kor, Propositions 2 and 7]. The strong convergence convention of [GKM, Sch] also requires the models to reproduce the operator norms of the left regular representation, so a group that is not MF as defined here is not MF in that convention either. Equivalently, G embeds in the unitary group of the norm matrix corona

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

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Here and below, products of C^*-algebras mean bounded products, and $\bigoplus$ denotes the ideal of norm-null sequences. The quotient $\mathcal{Q}_{\mathbf{d}}$ is the corona algebra of $\bigoplus_n M_{d_n}(\mathbb{C})$, since $\prod_n M_{d_n}(\mathbb{C})$ is its multiplier algebra. We call a homomorphism $G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ a *corona homomorphism*. A separable C^*-algebra is MF if it embeds in a norm matrix corona [BK].

The groups here fail to be MF because they contain a subgroup that is conjugated into itself. For $L \leq G$ put

$$\mathrm{Comp}_G(L) = \{u \in G : uLu^{-1} \leq L\}.$$

An element c of the centralizer $C_G(L)$ commutes with L, so ucu^{-1} commutes with uLu^{-1} but need not commute with the rest of L. The normal subgroup of G generated by the commutators of all such elements ucu^{-1} with elements of L is

$$\mathfrak{D}_G(L) = \langle\!\langle [ucu^{-1}, \ell] : u \in \mathrm{Comp}_G(L),\ c \in C_G(L),\ \ell \in L \rangle\!\rangle_G. \tag{1}$$

Theorem A (one-sided compression criterion)**.** *Let G be countable and let $L \leq G$ have property* (T)*. If $K \trianglelefteq G$ has property* (T) *and $K \leq \mathfrak{D}_G(L)$, then every homomorphism from G to an MF group is trivial on K. In particular, a nontrivial such K makes G non-MF, and if G has property* (T) *and $\mathfrak{D}_G(L) = G$, then every homomorphism from G to an MF group is trivial.*

Asymptotic multiplicativity in operator norm makes the conjugation maps $\mathrm{Ad}(V_n(g))\colon x \mapsto V_n(g)xV_n(g)^*$ an asymptotically multiplicative family of unitaries on $M_{d_n}(\mathbb{C})$ with its normalized Hilbert–Schmidt inner product. Here $\|\mathrm{Ad}(A) - \mathrm{Ad}(B)\| \leq 2\,\|A - B\|$, with the norm on the left taken in the operator algebra of this Hilbert space. Passing to a Hilbert ultraproduct turns these maps into a genuine unitary representation of G, to which property (T) applies. The unitary U of the compressor u satisfies $U^*PU \leq P$ for the Kazhdan projection P of L, because $uLu^{-1} \leq L$ has fewer constraints and more fixed vectors, and stable finiteness of the matrix corona forces $U^*PU = P$. So conjugation by u preserves the Hilbert–Schmidt asymptotic commutant of L, and every element of $\mathfrak{D}_G(L)$ tends to 1 in Hilbert–Schmidt norm in every operator norm asymptotic representation (Corollary 2.5). Property (T) of K turns this into triviality in operator norm: a corona homomorphism nontrivial on K compresses to the corner where K has no fixed vectors, and the Kazhdan inequality there contradicts the Hilbert–Schmidt triviality (Theorem 2.7).[1]

The stable letter of an ascending HNN extension of a property-(T) group along a proper self-embedding is a compressor. The group H realizes the compression inside a simple group, where one nontrivial commutator $[ucu^{-1}, \ell]$ normally generates everything. Let $L_{\mathbb{F}_2}(1,2)$ denote the binary Leavitt algebra, the unital $\mathbb{F}_2$-algebra generated by s_0, s_1, t_0, t_1 subject to

$$t_i s_j = \delta_{ij}, \qquad s_0 t_0 + s_1 t_1 = 1. \tag{2}$$

For a unital ring R, write $e_{ij}(a) = 1 + aE_{ij}$ and let $\mathrm{EL}_n(R)$ be the subgroup of $\mathrm{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \neq j$.

[1] A Hilbert–Schmidt bound on the multiplicative defect does not give an operator norm bound on the conjugation maps, so this argument does not apply to sofic or hyperlinear approximations.

Theorem B (the binary Leavitt group). *Put* $R = L_{\mathbb{F}_2}(1,2)$ *and* $H = \mathrm{EL}_{12}(R)$. *Then* H *is finitely generated, nontrivial, simple, has property* (T), *and every homomorphism from* H *to an MF group is trivial. So* H *is not MF, and* $C^*_{\mathrm{r}}(H)$ *is separable and stably finite but not MF, while* $C^*_{\max}(H)$ *is not even finite: it contains a proper isometry.*

Theorem C (a torsion-free finitely presented example). *There is a two-generated, finitely presented, torsion-free, acylindrically hyperbolic group* Q *with property* (T) *such that every homomorphism from* Q *to an MF group is trivial. In particular, no nontrivial quotient of* Q *is MF.*

Section 5 builds Q from Fournier-Facio's torsion-free property-(T) group [FF, §2] and Hull's small cancellation theorem [Hu].

We construct an affine–Clifford group W that is an extension of $\mathbb{Z}$ by a locally residually finite group but is not MF. Its canonical trace on $C^*_{\max}(W)$ is amenable and not quasidiagonal. This answers Brown's question whether every amenable trace is quasidiagonal [Br, discussion preceding Proposition 3.5.1].

Relation to prior work. Blackadar and Kirchberg introduced MF algebras and proved that a separable C^*-algebra is NF if and only if it is nuclear and MF [BK]. Whether every separable stably finite C^*-algebra is MF, the MF problem [GH, §6], was answered negatively through the failure of the Connes embedding problem [JNVWY][GH, Proposition 6.1 and Remark 6.2]. Theorem B gives a counterexample among reduced group C^*-algebras. Whether every countable group is MF was a longstanding open problem [BDL].

Theorem A uses the configuration of the nonsofic construction [OAI, Proposition 2.3]: a property-(T) subgroup L conjugated into itself by an element u, and an element c commuting with L whose conjugate ucu^{-1} does not. There, soficity and the rigidity theorems of Kun [Ku16] and Kun–Thom [KT19] force the centralizing subgroup to be locally embeddable into finite groups, and a copy of Thompson's group V gives the contradiction. Here the rigidity is the Kazhdan projection of L in a norm matrix corona, which stable finiteness of the corona forces to commute with the compressor, and the contradiction is a commutator $[ucu^{-1}, \ell] \neq 1$ that every operator norm asymptotic representation sends to 1 in Hilbert–Schmidt norm. The group H is isomorphic to the unit group of the binary Leavitt algebra, the group of [OAI], and Theorem C reuses Fournier-Facio's configuration [FF]. Bachner–Dogon–Lubotzky showed that for groups of Deligne type, operator–Hilbert–Schmidt stability would imply that the group is not MF [BDL, Proposition 1.5]. The compression relation replaces stability. For a short elementary construction of non-MF generalized wreath products from one-sided compression, see [Ec].

Leavitt introduced the algebras $L_K(1,n)$ [Lev]. The binary Leavitt algebra is purely infinite simple [AA], is an exchange ring [Ara], and has center $\mathbb{F}_2$ [AC]. Preusser's sandwich theorem then gives simplicity of H [Pre], and property (T) is Ershov–Jaikin-Zapirain's theorem [EJZ, Theorem 1.1].

2. One-sided compression

2.1. **Corona homomorphisms.** If G is countable, the image of a corona homomorphism from G is a countable subgroup of a corona unitary group, so MF. Every MF group embeds in the unitary group of a norm matrix corona.

Lemma 2.1. *Let G be countable and $K \le G$. Every homomorphism from G to an MF group is trivial on K if and only if every corona homomorphism from G is trivial on K.*

Proof. The image of a corona homomorphism is a countable MF group. Conversely, compose a homomorphism to an MF group with a corona embedding of its image; the resulting corona homomorphism has the same kernel. □

2.2. **Kazhdan transport in normalized Hilbert–Schmidt norm.** For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \operatorname{tr}_d(a^*a)^{1/2}, \qquad \operatorname{tr}_d = \frac{1}{d}\operatorname{Tr}.$$

An *operator norm asymptotic representation* of a group G is a sequence of maps $V_n \colon G \to \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G).$$

The elements g with $\|V_n(g) - 1\|_2 \to 0$ form a normal subgroup of G, because $\|a\|_2 \le \|a\|$, the Hilbert–Schmidt norm is invariant under adjoints and unitary conjugation, and $\|V_n(g^{-1}) - V_n(g)^*\| \to 0$:

$$\|V_n(gh) - 1\|_2 \le \|V_n(g) - 1\|_2 + \|V_n(h) - 1\|_2 + o(1),$$
$$\left\|V_n(g^{-1}) - 1\right\|_2 = \|V_n(g) - 1\|_2 + o(1), \qquad \left\|V_n(ghg^{-1}) - 1\right\|_2 = \|V_n(h) - 1\|_2 + o(1).$$

For $L \le G$, let

$$\mathcal{C}_2(V, L) = \left\{ (x_n) \;\middle|\; \begin{array}{l} \sup_n \|x_n\| < \infty, \\ \|V_n(\ell)x_n - x_nV_n(\ell)\|_2 \longrightarrow 0 \quad (\ell \in L) \end{array} \right\}$$

be the bounded Hilbert–Schmidt asymptotic commutant of $V(L)$. Coordinatewise conjugation by $V_n(g)$ defines a bijection, denoted $\operatorname{Ad}(V(g))$, of the bounded matrix sequences.

Lemma 2.2. *For every sequence of positive integers (m_n), the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C}) \Big/ \bigoplus_n M_{m_n}(\mathbb{C})$$

is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.

Proof. Write A for the displayed corona. If $v^*v = 1$ in A and (x_n) is a bounded lift of v, then $\|x_n^*x_n - 1\| \to 0$, so for all large n the matrix $u_n = x_n(x_n^*x_n)^{-1/2}$ is unitary and $\|u_n - x_n\| \to 0$; after assigning arbitrary unitary values to the finitely many remaining coordinates, (u_n) is a unitary lift of v, and $vv^* = 1$. So A is finite, and so is $M_k(A)$, which is the corona with matrix sizes km_n. Finally, if $p \le q$ are equivalent projections in a stably finite algebra B and w is a partial isometry with $w^*w = q$ and $ww^* = p$, then $w \in qBq$ is an isometry in the finite corner qBq, so $p = ww^* = q$. $\square$

Lemma 2.3 (one-sided order for the Kazhdan projection)**.** *Let L have property* (T)*, let B be a unital C^*-algebra, and let $\pi\colon L \to \mathcal{U}(B)$ be a homomorphism. Denote by $P \in B$ the image of the Kazhdan projection under the extension $C^*_{\max}(L) \to B$. If $U \in \mathcal{U}(B)$ satisfies*

$$U\pi(L)U^* \subseteq \pi(L),$$

then

$$U^*PU \le P.$$

Proof. Represent B faithfully and nondegenerately on a Hilbert space $\mathcal{H}$. Then P is the orthogonal projection onto $\operatorname{Fix}\pi(L)$. If ξ is L-fixed and $\ell \in L$, then $\pi(\ell)U^*\xi = U^*(U\pi(\ell)U^*)\xi = U^*\xi$ because $U\pi(\ell)U^*$ belongs to $\pi(L)$. So $U^* \operatorname{Fix}\pi(L) \subseteq \operatorname{Fix}\pi(L)$, and the range projection U^*PU is dominated by P in $B(\mathcal{H})$, and then in B by faithfulness. $\square$

Theorem 2.4 (one-sided Kazhdan transport)**.** *Let $L \le G$ have property* (T)*, and let $u \in G$ satisfy $uLu^{-1} \le L$. Let (V_n) be an operator norm asymptotic representation of G. Then*

$$\operatorname{Ad}(V(u))(\mathcal{C}_2(V,L)) = \mathcal{C}_2(V,L).$$

Proof. Fix $x = (x_n) \in \mathcal{C}_2(V,L)$ and a free ultrafilter ω, and let $\mathcal{K}_\omega$ be the ultraproduct of the Hilbert spaces $(M_{d_n}(\mathbb{C}), \|\cdot\|_2)$. For unitaries A, B,

$$\|\operatorname{Ad}(A) - \operatorname{Ad}(B)\| \le 2\,\|A - B\|\,,$$

where the norm on the left is the operator norm on this Hilbert space, so the maps $\operatorname{Ad}(V_n(g))$ are asymptotically multiplicative in operator norm, and

$$\sigma(g) = \lim_{\omega} \operatorname{Ad}(V_n(g))$$

is a unitary representation of G on $\mathcal{K}_\omega$ with $\xi = [x_n]_\omega \in \operatorname{Fix}\sigma(L)$. The same estimate makes

$$\widetilde{\sigma}(g) = [\operatorname{Ad}(V_n(g))]_n \in \mathcal{U}(\mathcal{B}), \qquad \mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C}))/\bigoplus_n B(M_{d_n}(\mathbb{C})),$$

a homomorphism, where $B(M_{d_n}(\mathbb{C}))$ denotes the operators on the Hilbert–Schmidt Hilbert space. Coordinatewise action induces a representation $\pi_\omega\colon \mathcal{B} \to B(\mathcal{K}_\omega)$, which we suppress from the notation. Moreover, $\mathcal{B}$ is a norm matrix corona with coordinate sizes d_n^2 after a choice of matrix units. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection of L [AW] under $C^*_{\max}(L) \to \mathcal{B}$. On $\mathcal{K}_\omega$ its range is $\operatorname{Fix}\sigma(L)$. Put $U = \widetilde{\sigma}(u)$. Since $uLu^{-1} \le L$,

$$U\widetilde{\sigma}(L)U^* \subseteq \widetilde{\sigma}(L), \qquad \text{so} \qquad U^*PU \le P$$

by Lemma 2.3, and

$$U^*PU = P$$

by Lemma 2.2, since the two projections are unitarily equivalent. So U commutes with P, and

$$U\xi,\ U^*\xi \in \operatorname{Fix}\sigma(L), \qquad \text{that is,}$$
$$\lim_{\omega} \left\| V_n(\ell)V_n(u)^{\pm 1}x_n V_n(u)^{\mp 1} - V_n(u)^{\pm 1}x_n V_n(u)^{\mp 1}V_n(\ell)\right\|_2 = 0 \qquad (\ell \in L).$$

Since ω was arbitrary, the limits hold along the full sequence: a subsequence staying above some $\varepsilon > 0$ would lie in a free ultrafilter and give a nonzero ultralimit. So $\operatorname{Ad}(V(u))$ and its inverse both preserve $\mathcal{C}_2(V, L)$. □

Corollary 2.5. *Let $L \le G$ have property* (T) *and let (V_n) be an operator norm asymptotic representation of G. Then*

$$\|V_n(d) - 1\|_2 \longrightarrow 0 \qquad (d \in \mathfrak{D}_G(L)).$$

Proof. Let $u \in \operatorname{Comp}_G(L)$, $c \in C_G(L)$, and $\ell \in L$, and write $\mathcal{C}_2 = \mathcal{C}_2(V, L)$. Then

$$(V_n(c)) \in \mathcal{C}_2 \implies (V_n(u)V_n(c)V_n(u)^*) \in \mathcal{C}_2 \implies (V_n(ucu^{-1})) \in \mathcal{C}_2,$$

the first membership because c commutes with L, the first implication by Theorem 2.4, and the second because $\|V_n(u)V_n(c)V_n(u)^* - V_n(ucu^{-1})\| \to 0$. So $\|V_n(\ell)V_n(ucu^{-1}) - V_n(ucu^{-1})V_n(\ell)\|_2 \to 0$. Hence, by asymptotic multiplicativity, $\|V_n([ucu^{-1}, \ell]) - 1\|_2 \to 0$. The elements with this property form a normal subgroup of G, so it contains $\mathfrak{D}_G(L)$. □

2.3. **From Hilbert–Schmidt to operator norm.** A corona homomorphism ρ cannot simply be restricted to a corner: a projection q commuting with $\rho(G)$ has lifts q_n that only asymptotically commute with the lifts of $\rho(g)$, so the compressions $q_n\rho(g)q_n$ are only approximately unitary.

Lemma 2.6 (central corona corners)**.** *Let $\rho\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ be a homomorphism from a countable group, and let $q \in \mathcal{Q}_{\mathbf{d}}$ be a nonzero projection commuting with $\rho(G)$. Then, after passing to an infinite coordinate subsequence, there are nonzero projections $q_n \in M_{d_n}(\mathbb{C})$, integers $r_n = \operatorname{rank}(q_n)$, unitary identifications $J_n\colon \mathbb{C}^{r_n} \to q_n\mathbb{C}^{d_n}$, and an operator norm asymptotic representation*

$$W_n\colon G \longrightarrow \mathcal{U}(r_n)$$

with $W_n(1) = I_{r_n}$ such that, in the corona over the retained coordinates, the class of $(J_nW_n(g)J_n^*)$ is the coordinate restriction of $q\rho(g)$, a unitary of the corner $q\mathcal{Q}_{\mathbf{d}}q$ with unit q.

Proof. Lift q to projections q_n by spectral rounding, and lift each $\rho(g)$ to unitaries $U_n(g)$ by polar correction, with $U_n(1) = I_{d_n}$. Since $q \ne 0$, infinitely many q_n are nonzero. Retain those coordinates and put $r_n = \operatorname{rank}(q_n)$. The commutation relation in the corona gives $\|q_nU_n(g) - U_n(g)q_n\| \to 0$ for each g, so [BDL, Proposition 2.4] with $p = \infty$ gives an operator norm asymptotic representation $\widetilde{W}_n$ of G by unitaries of the corner $q_nM_{d_n}(\mathbb{C})q_n$ with $\left\|\widetilde{W}_n(g) - q_nU_n(g)q_n\right\| \to 0$. Replacing $\widetilde{W}_n(1)$ by q_n changes it by $o(1)$. Choose unitaries $J_n\colon \mathbb{C}^{r_n} \to q_n\mathbb{C}^{d_n}$ and put $W_n(g) = J_n^*\widetilde{W}_n(g)J_n$. Conjugation by J_n preserves operator norms and multiplicative defects, so (W_n) is an operator norm asymptotic representation with $W_n(1) = I_{r_n}$, and $\|J_n W_n(g) J_n^* - q_n U_n(g) q_n\| \to 0$ identifies the class of $(J_n W_n(g) J_n^*)$ with the coordinate restriction of $q\rho(g)$. □

Theorem 2.7 (normal Kazhdan subgroups)**.** *Let G be countable and let $K \trianglelefteq G$ have property* (T)*. If every operator norm asymptotic representation (V_n) of G satisfies $\|V_n(k) - 1\|_2 \to 0$ for all $k \in K$, then every corona homomorphism from G is trivial on K.*

Proof. Suppose that a corona homomorphism $\Theta\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is nontrivial on K. Replace G by $\Theta(G)$ and K by $\Theta(K)$: the new K is a nontrivial normal property-(T) subgroup, since property (T) passes to quotients, and it still satisfies the hypothesis, because every operator norm asymptotic representation of $\Theta(G)$ pulls back along $G \to \Theta(G)$ to one of G.

Let $p \in \mathcal{Q}_{\mathbf{d}}$ be the image of the Kazhdan projection of K, the projection onto the K-fixed vectors in any faithful representation of the corona. Since $gKg^{-1} = K$,

$$\Theta(g) p \Theta(g)^* = p \qquad (g \in G),$$

and $q = 1 - p$ is nonzero, because $p = 1$ would make every element of K act trivially. Lemma 2.6 gives an operator norm asymptotic representation $W_n\colon G \to \mathcal{U}(r_n)$ whose corona homomorphism $\widehat{\Theta}$ is the coordinate restriction of $g \mapsto q\Theta(g)$. The Kazhdan projection of K under $\widehat{\Theta}$ is the coordinate restriction of $qp = 0$, so $\widehat{\Theta}|_K$ has no nonzero fixed vectors, and for a finite Kazhdan set $S \subseteq K$ with constant $\kappa > 0$,

$$b = \frac{1}{|S|} \sum_{s \in S} (\widehat{\Theta}(s) - 1)^* (\widehat{\Theta}(s) - 1) \;\geq\; \frac{\kappa^2}{|S|} 1$$

in $\mathcal{Q}_{\mathbf{r}}$. The coordinates $b_n = \frac{1}{|S|} \sum_{s \in S} (W_n(s) - I_{r_n})^* (W_n(s) - I_{r_n})$ represent b, so the negative part of $b_n - (\kappa^2/|S|) I_{r_n}$ tends to zero in operator norm, and taking normalized traces on $M_{r_n}(\mathbb{C})$ gives

$$\frac{1}{|S|} \sum_{s \in S} \|W_n(s) - I_{r_n}\|_2^2 \geq \frac{\kappa^2}{|S|} - o(1).$$

After passing to a subsequence, one fixed $s_0 \in S$ stays a positive Hilbert–Schmidt distance from I_{r_n}, against the hypothesis applied to (W_n) and $s_0 \in K$. □

Proof of Theorem A. By Corollary 2.5, every operator norm asymptotic representation of G satisfies $\|V_n(k) - 1\|_2 \to 0$ for $k \in K \leq \mathfrak{D}_G(L)$. Theorem 2.7 then makes every corona homomorphism trivial on K, and Lemma 2.1 makes every homomorphism to an MF group trivial on K. If $K \neq 1$, then G does not embed in an MF group, so G is not MF. The case $K = G = \mathfrak{D}_G(L)$ gives the last assertion. □

2.4. **The maximal group C^*-algebra.** Lemma 2.3 also applies inside $C^*_{\max}(G)$ itself, where a strict compression produces a proper isometry.

Lemma 2.8 (strictly dominated projections)**.** *Let A be a unital C^*-algebra containing a projection p and a unitary u with $p < upu^*$. Then $s = pu^* + (1 - upu^*)$ is an isometry that is not a unitary, so A is not stably finite and has no faithful tracial state.*

Proof. Put $q = upu^*$, so that $pq = qp = p$, $up(1-q) = 0$, and $pu^*(1-q) = pu^* - pu^*upu^* = 0$. Then

$$\begin{aligned} s^*s &= upu^* + up(1-q) + (1-q)pu^* + (1-q) = q + (1-q) = 1, \\ ss^* &= pu^*up + pu^*(1-q) + (1-q)up + (1-q) \\ &= p + (1-q) = 1 - (q-p) \neq 1. \end{aligned}$$

A unital C^*-algebra containing a nonunitary isometry is not finite, and neither is any matrix algebra over it, since $\mathrm{diag}(s, 1, \ldots, 1)$ is a nonunitary isometry in $M_k(A)$. A tracial state τ has $\tau(q-p) = \tau(s^*s) - \tau(ss^*) = 0$ with $q - p$ a nonzero positive element, so no tracial state on A is faithful. □

Proposition 2.9 (proper one-sided compression)**.** *Let G be a countable group containing a property-*(T) *subgroup Γ and an element t with $t\Gamma t^{-1} \subsetneq \Gamma$. Then $C^*_{\max}(G)$ contains a proper isometry. In particular, $C^*_{\max}(G)$ is not stably finite, admits no faithful tracial state, and is neither residually finite-dimensional nor MF.*

Proof. Let $P \in C^*_{\max}(G)$ be the image of the Kazhdan projection of Γ [AW] under the canonical map $C^*_{\max}(\Gamma) \to C^*_{\max}(G)$, and let u be the canonical unitary of t. Conjugation by u implements the isomorphism $\Gamma \to t\Gamma t^{-1}$ on canonical unitaries, so Lemma 2.3, applied with $U = u$, gives $P \leq uPu^*$ with uPu^* the image of the Kazhdan projection of $t\Gamma t^{-1}$. The inequality is strict: in the quasi-regular representation of G on $\ell^2(G/t\Gamma t^{-1})$, the point mass at the base coset is fixed by $t\Gamma t^{-1}$ and moved by every element of $\Gamma \setminus t\Gamma t^{-1}$, so the two fixed-space projections differ. Lemma 2.8 then supplies the proper isometry and rules out stable finiteness and faithful traces. MF algebras are stably finite, and a separable residually finite-dimensional C^*-algebra is MF [BK]. □

3. The binary Leavitt group

The Leavitt algebra $R = L_{\mathbb{F}_2}(1,2)$ is built so that one copy of the free module R is isomorphic to two copies: the relations (2) say that s_0, s_1 embed two copies of R side by side and t_0, t_1 read them back. So a matrix A over R can be copied into one of the two halves, with the identity on the other half, and this compression is an injective multiplicative map that fixes the identity. Its restriction to the unit group is an injective group homomorphism. Inside a large enough elementary group the compression is realized by conjugation, which gives a compressor τ for a copy L of $\mathrm{EL}_3(R)$; one commutator computation gives a nontrivial element of $\mathfrak{D}_H(L)$, and simplicity of H turns it into all of H.

Throughout this section $R = L_{\mathbb{F}_2}(1,2)$ and

$$p = s_0 t_0, \qquad q = s_1 t_1,$$

so that $p + q = 1$ and $t_1 q s_1 = 1$, and in particular $q \neq 0$. We use indices $0, \ldots, 11$ for 12×12 matrices and identify $\mathrm{GL}_3(R)$ with the subgroup of $\mathrm{GL}_{12}(R)$ of matrices $\mathrm{diag}(A, I_9)$. Define $\Psi\colon M_3(R) \to M_3(R)$ by

$$\Psi(A) = qI_3 + s_0 A t_0, \tag{3}$$

where $s_0 A t_0$ is the matrix $(s_0 a_{ij} t_0)_{ij}$. The relations $q^2 = q$, $qs_0 = t_0 q = 0$, and $t_0 s_0 = 1$ show that Ψ is multiplicative, $\Psi(I_3) = (p+q)I_3 = I_3$, and $t_0 \Psi(A) s_0 = A$. Thus it restricts to a group homomorphism $\mathrm{GL}_3(R) \to \mathrm{GL}_3(R)$, and the recovery identity makes that homomorphism injective. Put

$$X = \begin{pmatrix} s_0 I_3 & s_1 t_0 I_3 \\ 0 & t_1 I_3 \end{pmatrix}, \qquad Y = \begin{pmatrix} t_0 I_3 & 0 \\ s_0 t_1 I_3 & s_1 I_3 \end{pmatrix};$$

block multiplication with (2) gives $Y = X^{-1}$ and $X \operatorname{diag}(A, I_3) Y = \operatorname{diag}(\Psi(A), I_3)$. The matrix X need not lie in an elementary group, which is why there are twelve coordinates: with

$$\tau = \operatorname{diag}(X, Y) \in \mathrm{GL}_{12}(R) \tag{4}$$

one has

$$\tau \operatorname{diag}(A, I_9)\, \tau^{-1} = \operatorname{diag}(\Psi(A), I_9). \tag{5}$$

Lemma 3.1. *The matrix τ defined in (4) belongs to $\mathrm{EL}_{12}(R)$.*

Proof. By the Whitehead lemma [Mil, §3], $\operatorname{diag}(X, X^{-1})$ is a product of block-triangular matrices $\left(\begin{smallmatrix} I & B \\ 0 & I \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} I & 0 \\ B & I \end{smallmatrix}\right)$ with $B \in M_6(R)$, and each of these is the product of the elementary matrices $e_{r,6+s}(b_{rs})$, respectively $e_{6+r,s}(b_{rs})$, since distinct off-diagonal matrix units have product zero. $\square$

Set $H = \mathrm{EL}_{12}(R)$, and let $L = \mathrm{EL}_3(R) \le H$ denote the image of the embedding $A \mapsto \operatorname{diag}(A, I_9)$.

Proposition 3.2. *The groups H and L have property* (T)*, and $\tau \in H$ satisfies*

$$\tau L \tau^{-1} \le L. \tag{6}$$

Proof. The ring R is generated by s_0, s_1, t_0, t_1 as a unital ring, so the theorem of Ershov and Jaikin-Zapirain gives property (T) for $\mathrm{EL}_{12}(R)$ and $\mathrm{EL}_3(R)$ [EJZ, Theorem 1.1]. Lemma 3.1 gives $\tau \in H$. For $i \neq j$ and $a \in R$, equations (5) and (3) give $\tau e_{ij}(a) \tau^{-1} = e_{ij}(s_0 a t_0) \in L$, and these elementary matrices generate L. $\square$

Proposition 3.3. *The group $H = \mathrm{EL}_{12}(R)$ is nontrivial and simple.*

Proof. The ring R is purely infinite simple [AA], so it is an exchange ring [Ara]. Let $N \trianglelefteq H$. Since $N \le \mathrm{GL}_{12}(R)$ is normalized by $\mathrm{EL}_{12}(R)$, Preusser's sandwich theorem [Pre, Theorem 3] gives an ideal $I \trianglelefteq R$ with $\mathrm{EL}_{12}(R, I) \le N \le C_{12}(R, I)$, where $\mathrm{EL}_{12}(R, I)$ is the relative elementary subgroup and $C_{12}(R, I)$ is the full congruence subgroup. Simplicity of R gives $I = 0$ or $I = R$, and $I = R$ gives $N = H$. If $I = 0$, then N centralizes every $e_{ij}(a)$: centralizing $e_{ij}(1)$ for all $i \neq j$ forces $g = \lambda I_{12}$ with $\lambda \in R^\times$, and centralizing $e_{ij}(a)$ forces $\lambda \in Z(R)^\times$. Since $Z(R) = \mathbb{F}_2$ [AC, Corollary 4.3], $N = 1$. Finally, $e_{01}(1) \neq 1$. $\square$

Proposition 3.4. *Let*

$$c = e_{34}(1), \qquad \ell = e_{12}(1), \qquad d = [\tau c \tau^{-1}, \ell]$$

in H. Then

$$c \in C_H(L), \qquad d = e_{02}(q) \neq 1, \qquad \langle\!\langle d \rangle\!\rangle_H = H.$$

Proof. The Steinberg relations give $[e_{ij}(a), e_{34}(1)] = 1$ for $i, j \in \{0,1,2\}$, and these matrices generate L, so $c \in C_H(L)$. Since $\tau = \mathrm{diag}(X, Y)$ and E_{34} is supported in the first six coordinates, $\tau E_{34}\tau^{-1} = \mathrm{diag}(XE_{34}Y, 0)$ with E_{34} read in $M_6(R)$, and with $\varepsilon_0, \varepsilon_1, \varepsilon_2$ the standard basis of R^3,

$$XE_{34}Y = \begin{pmatrix} s_1t_0\varepsilon_0 \\ t_1\varepsilon_0 \end{pmatrix} \begin{pmatrix} s_0t_1\varepsilon_1^{\mathsf{T}} & s_1\varepsilon_1^{\mathsf{T}} \end{pmatrix} = \begin{pmatrix} s_1t_1E_{01} & 0 \\ 0 & E_{01} \end{pmatrix}$$

by (2). So

$$\tau c\tau^{-1} = 1 + qE_{01} + E_{34} = e_{01}(q)e_{34}(1).$$

The second factor commutes with ℓ, and $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$ gives $d = e_{02}(q)$, which is nontrivial because $q \neq 0$. Since H is simple, d normally generates H. □

Proof of Theorem B. Proposition 3.2 gives property (T), which implies finite generation [BHV, Theorem 1.3.1]. The containment (6), the centrality of c relative to L, and Proposition 3.4 put d in $\mathfrak{D}_H(L)$, and $\langle\langle d\rangle\rangle_H = H$ gives $\mathfrak{D}_H(L) = H$. So Theorem A, applied with $K = H$, makes every homomorphism from H to an MF group trivial, and Proposition 3.3 makes H nontrivial and simple. In particular H is not MF. The algebra $C^*_{\mathrm{r}}(H)$ is separable, and stably finite because its canonical trace is faithful. If an embedding $e\colon C^*_{\mathrm{r}}(H) \to \mathcal{Q}_{\mathbf{d}}$ existed and $p = e(1)$, then $h \mapsto e(\lambda_h) + (1-p)$ would embed H in $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$, contradicting that H is not MF. Finally, the compression (6) is strict: by (5) and (3), every off-diagonal entry of a matrix in $\tau L\tau^{-1}$ has the form s_0at_0, and

$$t_1(s_0at_0)s_1 = 0 \neq 1 = t_1 \cdot 1 \cdot s_1,$$

so $e_{12}(1) \in L \setminus \tau L\tau^{-1}$. Proposition 2.9, applied to $L \leq H$ and τ, puts a proper isometry in $C^*_{\max}(H)$. □

4. An amenable nonquasidiagonal trace

Brown observed that it was unknown whether every amenable trace on a C^*-algebra is quasidiagonal [Br, discussion preceding Proposition 3.5.1]. We use the following sequential notions for an arbitrary unital C^*-algebra A. A trace τ is *amenable* if there are u.c.p. maps $\phi_n\colon A \to M_{d_n}(\mathbb{C})$ such that

$$\|\phi_n(ab) - \phi_n(a)\phi_n(b)\|_2 \longrightarrow 0, \qquad \mathrm{tr}_{d_n}(\phi_n(a)) \longrightarrow \tau(a) \quad (a, b \in A).$$

It is *quasidiagonal* if the first limit holds in operator norm instead of normalized Hilbert–Schmidt norm, with the same trace convergence.

For a group G, let u_g denote the canonical unitary in $C^*_{\max}(G)$. The canonical trace τ_G is determined by

$$\tau_G(u_g) = \begin{cases} 1, & g = 1, \\ 0, & g \neq 1. \end{cases}$$

Theorem 4.1 (canonical-trace obstruction)**.** *Let G be a group. If G is not MF, then its canonical trace τ_G on $C^*_{\max}(G)$ is not quasidiagonal. Consequently, if τ_G is amenable, then it is amenable and not quasidiagonal.*

Proof. Suppose that τ_G is quasidiagonal, and let ϕ_n be u.c.p. models as above. Operator-norm multiplicativity makes both $\phi_n(u_g)^*\phi_n(u_g)$ and $\phi_n(u_g)\phi_n(u_g)^*$ converge to 1. Polar correction therefore gives unitaries $V_n(g)$ with

$$\|V_n(g) - \phi_n(u_g)\| \longrightarrow 0.$$

The maps $V_n\colon G \to \mathcal{U}(d_n)$ are asymptotically multiplicative in operator norm. If $g \neq 1$, then $\mathrm{tr}_{d_n}(\phi_n(u_g)) \to \tau_G(u_g) = 0$. Hence $V_n(g)$ cannot converge to 1 in operator norm, since that would force $\mathrm{tr}_{d_n}(\phi_n(u_g)) \to 1$. Thus (V_n) is an MF approximation of G, contrary to the hypothesis. The final assertion merely combines this obstruction with amenability. □

A group N is *locally residually finite* if each of its finitely generated subgroups is residually finite.

Proposition 4.2 (extensions of $\mathbb{Z}$ by locally residually finite groups)**.** *Suppose that G is countable and*

$$G \cong N \rtimes_\beta \mathbb{Z},$$

where N is locally residually finite. Then τ_G is amenable.

Proof. Write an element of $N \rtimes_\beta \mathbb{Z}$ as (x, k) and call k its height, and write $t = (1, 1)$. Fix a finite set $F \subseteq G$ and the finite level window $I_L = \{0, \dots, L-1\}$. Moving the elements of F past the powers t^j for $j \in I_L$ produces finitely many normal coordinates. Let $N_{F,L} \leq N$ be the subgroup that they generate. It is residually finite, so there is a finite quotient

$$\theta_{F,L}\colon N_{F,L} \longrightarrow Q_{F,L}$$

that separates every nonidentity coordinate in this finite collection. Let $H_{F,L} \leq G$ be the image of $\ker \theta_{F,L} \leq N_{F,L}$ under the inclusion $N \hookrightarrow G$. Choose a representative $r_q \in N_{F,L}$ for each $q \in Q_{F,L}$ and set

$$S_{F,L} = \{t^j r_q H_{F,L} : j \in I_L,\ q \in Q_{F,L}\} \subseteq G/H_{F,L}.$$

These cosets are pairwise distinct. Let $\lambda_{F,L}$ be the quasi-regular representation of G on $\ell^2(G/H_{F,L})$ and let $P_{F,L}$ be the projection onto $\ell^2(S_{F,L})$. Put

$$\phi_{F,L}(b) = P_{F,L}\lambda_{F,L}(b)P_{F,L}\big|_{\ell^2(S_{F,L})} \qquad (b \in C^*_{\max}(G)).$$

This map is u.c.p., being the compression of a unitary representation.

Let u_g denote the canonical unitary of g. For $g, h \in F$, the two operators $\phi_{F,L}(u_{gh})$ and $\phi_{F,L}(u_g)\phi_{F,L}(u_h)$ agree on the basis vector indexed by $s \in S_{F,L}$ whenever $h \cdot s$ remains in the window. The escaping vectors lie over the boundary levels

$$\partial_{k(h)} I_L = \{j \in I_L : j + k(h) \notin I_L\}.$$

The quotient coordinate contributes the same multiplicity above every level, and therefore

$$\|\phi_{F,L}(u_{gh}) - \phi_{F,L}(u_g)\phi_{F,L}(u_h)\|_2^2 \leq \frac{|\partial_{k(h)} I_L|}{|I_L|}.$$

For fixed h the right-hand side tends to zero as $L \to \infty$.

The normalized trace of $\phi_{F,L}(u_g)$ counts the points of the finite window fixed by g. If g fixed $wH_{F,L}$, then $w^{-1}gw \in H_{F,L} \leq N$, and hence $k(g) = 0$. Thus a nonzero-height element fixes no selected coset. If $k(g)=0$ and $g\neq 1$, fixing the coset at level j would put its nontrivial normal coordinate at that level in $\ker\theta_{F,L}$, contrary to the choice of the quotient. Thus

$$\mathrm{tr}(\phi_{F,L}(u_g))=0 \quad (g\in F\setminus\{1\}), \qquad \mathrm{tr}(\phi_{F,L}(1))=1.$$

Take an exhaustion $F_1\subseteq F_2\subseteq\cdots$ of G and a diagonal choice $L_n\to\infty$. The maps ϕ_{F_n,L_n} are asymptotically multiplicative in normalized Hilbert–Schmidt norm and their traces converge to τ_G on every canonical unitary. The linear span of these unitaries is dense in $C^*_{\max}(G)$; contractivity extends both limits to the whole algebra, which is precisely amenability of τ_G. □

The obstruction itself only needs a proper self-embedding of a Kazhdan group. Let Γ be a countable group with property (T), let $\alpha\colon\Gamma\to\Gamma$ be injective but not surjective, and choose $a\in\Gamma\setminus\alpha(\Gamma)$. Put

$$T_\alpha=\varinjlim(\Gamma\xrightarrow{\alpha}\Gamma\xrightarrow{\alpha}\cdots), \qquad V=T_\alpha\rtimes\langle t\rangle,$$

where $tgt^{-1}=\alpha(g)$ on the level-zero copy of Γ. Put $X=V/\Gamma$, the left-coset space. The Clifford lamp group $\mathrm{Cl}(X)$ is presented by generators ε and c_x for $x\in X$, with relations

$$\varepsilon^2=c_x^2=1, \qquad [\varepsilon,c_x]=1, \qquad [c_x,c_y]=\varepsilon \quad (x\neq y).$$

The permutation action of V on X acts on these lamps, and the affine–Clifford witness is

$$W=\mathrm{Cl}(X)\rtimes V. \tag{7}$$

Proposition 4.3 (Clifford obstruction from a proper self-embedding)**.** *The group W is not MF. More precisely, every homomorphism from W to an MF group kills ε.*

Proof. Let c be the lamp at the root coset. It centralizes the level-zero copy of Γ. The two cosets supporting tct^{-1} and $a(tct^{-1})a^{-1}$ are distinct precisely because $a\notin\alpha(\Gamma)$, and therefore

$$[tct^{-1},a(tct^{-1})a^{-1}]=\varepsilon\neq 1. \tag{8}$$

Here and below the copy of Γ in W is understood. Set

$$x=tct^{-1}, \qquad y=axa^{-1}, \qquad d=[x,a].$$

Both x and y are involutions, and hence

$$d=xaxa^{-1}=xy, \qquad d^2=(xy)^2=[x,y]=\varepsilon.$$

Since $t\in\mathrm{Comp}_W(\Gamma)$, $c\in C_W(\Gamma)$ and $a\in\Gamma$, the element $d=[tct^{-1},a]$ is one of the generators of $\mathfrak{D}_W(\Gamma)$. Thus

$$\langle\varepsilon\rangle\leq\mathfrak{D}_W(\Gamma).$$

The subgroup $\langle\varepsilon\rangle$ is finite, central, normal and nontrivial, so it has property (T). The one-sided compression criterion applies because W is countable: the groups T_α and V, the coset space X, and $\mathrm{Cl}(X)$, whose elements have finite support, are all countable. It therefore makes every homomorphism from W to an MF group kill ε. In particular, W is not MF. □

The extra hypotheses in the next proposition concern approximation of the canonical trace, not the Clifford obstruction.

Proposition 4.4 (the locally residually finite case)**.** *Suppose in addition that* Γ *is residually finite and that* $[\Gamma : \alpha(\Gamma)] < \infty$*. Then*

$$W \cong K \rtimes \mathbb{Z}, \qquad K = \mathrm{Cl}(X) \rtimes T_\alpha,$$

where K *is locally residually finite. Consequently the canonical trace on* $C^*_{\max}(W)$ *is amenable and not quasidiagonal.*

Proof. Reassociating the semidirect products yields

$$W \cong K \rtimes \mathbb{Z}, \qquad K = \mathrm{Cl}(X) \rtimes T_\alpha.$$

The group K is locally residually finite . For $n \geq 0$, write $\Gamma_n = t^{-n}\Gamma t^n$ for the image of the nth copy of Γ in T_α. Successive telescope copies have finite relative index because $[\Gamma : \alpha(\Gamma)] < \infty$, so all the Γ_n are commensurable. In fact Γ is commensurated by V: every element of T_α lies in some telescope copy, and conjugation by t shifts the copies. Hence the stabilizer in Γ_n of every point $v\Gamma \in V/\Gamma$ has finite index, and every Γ_n-orbit in X is finite.

Every finite subset of K involves finitely many Clifford generators and has its T_α-coordinates in one Γ_n. The union Y of the Γ_n-orbits of the corresponding sites is finite and invariant, and the original finite subset is contained in

$$C_Y \rtimes \Gamma_n,$$

where $C_Y = \langle \varepsilon, c_y : y \in Y\rangle$ is finite by the displayed presentation of $\mathrm{Cl}(X)$. This semidirect product is residually finite. Indeed, an element with nontrivial Γ_n-component survives in a finite quotient of Γ_n. For a nontrivial element of C_Y, let J be the kernel of the action $\Gamma_n \to \mathrm{Aut}(C_Y)$. The quotient Γ_n/J is finite, and the finite semidirect product $C_Y \rtimes (\Gamma_n/J)$ retains that element. Therefore every finitely generated subgroup of K is residually finite.

Proposition 4.2 now makes the canonical trace amenable, and Theorem 4.1 makes it nonquasidiagonal. ☐

For a concrete instance, use the following six integral affine matrices:

$$x = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad y = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad z = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ -1 & -1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

$$v_1 = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

They generate

$$\bar{\Gamma} = \langle x, y, z, v_1, v_2, v_3\rangle = \left\{ \begin{pmatrix} A & v \\ 0 & 1 \end{pmatrix} : A \in \mathrm{SL}_3(\mathbb{Z}),\ v \in \mathbb{Z}^3 \right\} \cong \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z}).$$

It is residually finite and has property (T). The first assertion follows because reduction modulo a suitable integer separates any two distinct integral matrices. The second is the standard property (T) theorem for $\mathbb{Z}^n \rtimes \mathrm{SL}_n(\mathbb{Z})$, $n \geq 3$ [BHV, Example 1.7.4(i)].

Let
$$D = \mathrm{diag}(2, 2, 2, 1), \qquad \alpha(g) = DgD^{-1}.$$
Equivalently, $\alpha(v, A) = (2v, A)$. This is injective and its image consists exactly of the affine matrices with even translation coordinate, so $[\bar{\Gamma} : \alpha(\bar{\Gamma})] = 8$. If a is translation by the first standard basis vector, then $a \notin \alpha(\bar{\Gamma})$; equivalently, $D^{-1}aD$ has a $1/2$ entry. Hence the preceding construction applies to this $\bar{\Gamma}$, α and a.

Corollary 4.5 (the affine–Clifford trace)**.** *The canonical trace τ_W on $C^*_{\max}(W)$ is amenable but not quasidiagonal.*

Proof. The decomposition $W \cong K \rtimes \mathbb{Z}$ and local residual finiteness of K imply that τ_W is amenable by Proposition 4.2. Since W is not MF, Theorem 4.1 implies that τ_W is not quasidiagonal. □

5. A torsion-free finitely presented example

For an acylindrically hyperbolic group G, choose the generating set A provided by Hull [Hu, Theorem 3.12]. A subgroup is *suitable* if it acts non-elementarily on $\Gamma(G, A)$ and normalizes no nontrivial finite subgroup [Hu, Definition 1.4]. Hull's small cancellation theorem [Hu, Theorem 7.1] is stated with injectivity on a ball of $\Gamma(G, A)$, and a ball containing a given finite set gives the following form.

Theorem 5.1 (Hull's small cancellation theorem)**.** *Let G be acylindrically hyperbolic, let $N \leq G$ be suitable with respect to A, and let $g_1, \ldots, g_m \in G$ and a finite subset $\Omega \subseteq G$ be given. Then there is a surjective homomorphism $\varphi\colon G \to Q$ such that Q is acylindrically hyperbolic, $\varphi|_\Omega$ is injective, $\varphi(g_i) \in \varphi(N)$ for all i, and every element of finite order in Q is the image of an element of the same order in G.*

Hull's proof treats $m = 1$ by passing to $G/\langle\langle W \rangle\rangle_G$ for one word W and the general case by induction on m [Hu, proof of Theorem 7.1], so $\ker\varphi$ is the normal closure of m elements and Q is finitely presented when G is.

Lemma 5.2 (saturation)**.** *Let G be finitely presented, torsion-free, and acylindrically hyperbolic, let $N \trianglelefteq G$ be nontrivial, and let $\Omega \subseteq G$ be finite. Then there is a surjective homomorphism $\varphi\colon G \to Q$ such that Q is two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, $\varphi|_\Omega$ is injective, and $\varphi(N) = Q$.*

Proof. The subgroup N is infinite and normal, so it acts non-elementarily on $\Gamma(G, A)$ by Osin [Os, Lemma 7.1], and torsion-freeness makes it suitable. By Hull [Hu, Corollary 5.7 and Lemma 5.8], N contains two elements h_1, h_2 such that $N_0 = \langle h_1, h_2 \rangle$ is again suitable. Apply Theorem 5.1 to N_0 with $g_1, \ldots, g_m$ a finite generating set of G and with the set Ω. Then $Q = \langle \varphi(g_1), \ldots, \varphi(g_m) \rangle \leq \varphi(N_0) \leq \varphi(N) \leq Q$, so $Q = \varphi(N_0) = \langle \varphi(h_1), \varphi(h_2) \rangle$ and $\varphi(N) = Q$. The quotient is torsion-free by Theorem 5.1 and finitely presented because $\ker\varphi$ is normally generated by m elements. □

Fournier-Facio constructs a finitely presented torsion-free group G_0 with property (T), a subgroup $\Gamma \leq G_0$ with property (T), an element $t \in G_0$ with $t\Gamma t^{-1} \leq \Gamma$, and a subgroup $J \leq G_0$ isomorphic to a finitely presented infinite simple group, such that $[\Gamma, J] = 1$ and $tJt^{-1} \leq \Gamma$ [FF, §2]. The group G_0 is obtained there from Hull's common quotient theorem [Hu, Corollary 7.4], so it is acylindrically hyperbolic. Put $S = tJt^{-1}$. Since J is simple and nonabelian, J and S are perfect.

Lemma 5.3. *For every homomorphism $\rho\colon G_0 \to \bar{G}$,*

$$\rho(S) \leq \mathfrak{D}_{\bar{G}}(\rho(\Gamma)).$$

Proof. Write $\bar{\Gamma} = \rho(\Gamma)$, $\bar{t} = \rho(t)$, $\bar{J} = \rho(J)$, and $\bar{S} = \rho(S) = \bar{t}\bar{J}\bar{t}^{-1}$. Then $\bar{t}\bar{\Gamma}\bar{t}^{-1} \leq \bar{\Gamma}$, $\bar{J}$ centralizes $\bar{\Gamma}$, and $\bar{S} \leq \bar{\Gamma}$. For $c \in \bar{J}$ and $\ell \in \bar{S}$, the commutator $[\bar{t}c\bar{t}^{-1}, \ell]$ lies in $\mathfrak{D}_{\bar{G}}(\bar{\Gamma})$ by (1). As c ranges over $\bar{J}$, the element $\bar{t}c\bar{t}^{-1}$ ranges over $\bar{S}$, so $[\bar{S}, \bar{S}] \leq \mathfrak{D}_{\bar{G}}(\bar{\Gamma})$. The group $\bar{S}$ is perfect as a quotient of S, so $\bar{S} \leq \mathfrak{D}_{\bar{G}}(\bar{\Gamma})$. $\square$

Proof of Theorem C. Let N be the normal closure of S in G_0, and fix $s \in S$ with $s \neq 1$. Lemma 5.2 applied to G_0, N, and the finite set $\{1, s\}$ gives $\varphi\colon G_0 \to Q$ with Q two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, $\varphi(N) = Q$, and $\varphi(s) \neq 1$. The group Q has property (T) as a quotient of G_0.

Let $r\colon Q \to M$ be a homomorphism to an MF group with image $\bar{G}$, and put $\rho = r \circ \varphi$. Since $\varphi(N) = Q$, the normal closure of $\rho(S)$ in $\bar{G}$ is $\rho(N) = \bar{G}$, and by Lemma 5.3 it lies in the normal subgroup $\mathfrak{D}_{\bar{G}}(\rho(\Gamma))$, so $\mathfrak{D}_{\bar{G}}(\rho(\Gamma)) = \bar{G}$. The group $\bar{G}$ has property (T) as a quotient of Q, and $\rho(\Gamma)$ has property (T) as a quotient of Γ. Theorem A, applied to $\bar{G}$ with the subgroup $\rho(\Gamma)$ and $K = \bar{G}$, makes every homomorphism from $\bar{G}$ to an MF group trivial. But $\bar{G}$ is a subgroup of the MF group M, so it is MF, and its identity map is such a homomorphism. So $\bar{G} = 1$, and r is trivial. A homomorphism from a quotient of Q composes to one from Q, so no nontrivial quotient of Q is MF. $\square$

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